

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	1-3 Semester (2025-26)
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Branch	CSE
Course Outcome	5

Case Study Topic: Spam-Filter Feature Vector Sorting using In-Place Merge Sort

Work Done Summary:

In this case study, Merge Sort was used to sort feature vectors in a spam-filter training pipeline.

The training dataset contains large volumes of records that must be sorted efficiently while minimizing memory usage.

Implementation Details:

1. Recursive divide phase.
2. Merge phase.
3. Final sorted output.
4. Complexity analysis.
5. Stability discussion.
6. Comparison between standard and in-place merge sort.

Result:

Input :

[7,3,11,5,14,2,9,6,12,1,8,4]

Sorted Output :

[1,2,3,4,5,6,7,8,9,11,12,14]

Output:

[1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 14]

GitHub Link:

<https://github.com/saisharan134/DSA2-Case-Studies>

Student Signature	Faculty Signature